Aayam Regmi

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RESEARCH INTERESTS

Computer vision, robotic perception, and vision-based human–computer interaction — with a focus on pose estimation, real-time sensor fusion, and deep learning systems applied to physical-world environments. Seeking to pursue graduate research at the intersection of robotics and computer vision.

EDUCATION

B.Sc. (Hons) Computer Science — Database Specialization *Oct 2022 – May 2026 (Results Pending)*
Leeds Beckett University — via The British College, Kathmandu, Nepal (UK-accredited)

- GPA: ~3.9 / 4.0 est. (82–91% across all semesters; final results pending)
- Top 5% of graduating cohort
- 91% — 7th Semester; Merit / Refund Scholarship candidate
- Selected for International Student Exchange Programme (1 of ~200 students)

A Levels *Sep 2020 – May 2022*
Trinity International College — Kathmandu, Nepal

Secondary Education Examination (SEE) *Apr 2012 – Apr 2020*
Galaxy Public School — Bhaktapur, Nepal

RESEARCH

Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction *2026*
Lead Researcher — The British College / Leeds Beckett University

- Benchmarked MediaPipe and OpenPose (Convolutional Pose Machines) for real-time piano key-press detection via pose estimation and geometric containment.
- Designed and built a custom MIDI–video synchronisation pipeline; collected a dataset of 38 sessions comprising 4,777 annotated note-on events with Fitzpatrick skin-type, hand-size, and illuminance metadata.
- Proposed horizontal-axis Mean Per Joint Position Error (MPJPE) as a domain-specific localization metric to evaluate lateral fingertip precision.
- Achieved 63.3% polygon hit rate with MediaPipe vs 8.1% with OpenPose; identified a left–right hand detection asymmetry (95.8% right vs 53.2% left) informing future dataset balance requirements.
- Manuscript in preparation for conference submission.

PROJECTS

Tree Conservation Monitoring System *2025*
International Student Exchange Programme — 1st Place (Winning Team)

- Designed and assembled a custom battery-powered IoT device around an Arduino Mega ADK integrating a NEO-6M GPS module, DollaTek UART 2D barcode scanner, I2C SSD1306 OLED display, MicroSD card reader, and a 4-key input board over UART, I2C, and SPI buses.
- Device captures GPS coordinates, timestamps, and field metrics per tree; logs records to MicroSD for offline resilience and syncs to a central database for persistent tree-level history.
- Deployed physical barcodes on trees in the field; scanning a barcode with the handheld device instantly retrieves that tree's full record on the OLED display — no network required at scan time.
- **Technologies:** Arduino (C++), NEO-6M GPS, UART/I2C/SPI protocols, MicroSD storage, OLED display, barcode scanning, relational database.

Credit Card Default Prediction *2024*
Machine Learning Project

- Built and compared classification models (Random Forest, Logistic Regression, SVM, Gradient Boosting) on the UCI credit card default dataset; achieved 92% accuracy with Random Forest.
- Applied feature engineering, SMOTE oversampling for class imbalance, and cross-validated hyperparameter tuning.
- **Technologies:** Python, scikit-learn, Pandas, NumPy, Matplotlib.

Additional projects available at: github.com/AayamRegmi

WORK EXPERIENCE

Full-Stack Software Developer Intern — Professional Computer Systems, Kathmandu, Nepal *Nov 2024 – Mar 2025*

- Developed RESTful backend services and authentication flows using Spring Boot and Spring Security with a PostgreSQL database.
- Built responsive frontend components in ReactJS; collaborated on production deployment pipeline.

TECHNICAL SKILLS

Computer Vision & ML:	PyTorch, TensorFlow, OpenCV, MediaPipe, OpenPose, scikit-learn, Convolutional Neural Networks (CNN), pose estimation
Data Science:	NumPy, Pandas, Matplotlib, SciPy, Jupyter
Programming Languages:	Python (advanced), Java, JavaScript, SQL, PHP, Lua
Web & Backend:	Spring Boot, Spring Security, React, Node.js, Django, REST APIs
Databases:	PostgreSQL, Oracle, PL/SQL
Embedded & Hardware:	Arduino (C++), NEO-6M GPS, UART / I2C / SPI, OLED displays, MicroSD, barcode scanners
Tools & Environments:	Git, Linux/Ubuntu, Google Colab, Jupyter

AWARDS & HONORS

- 1st Place – International Student Exchange Programme Project Competition (2025)
- International Student Exchange Programme Selectee – chosen from 200+ applicants (2025)
- Top 5% of Graduating Cohort – Leeds Beckett University / The British College (2026)
- Merit / Refund Scholarship Candidate – 91% score, 7th Semester (2026)

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — *Coursera / Stanford & DeepLearning.AI (2023)*
- Advanced Learning Algorithms — *Coursera / Stanford & DeepLearning.AI (2023)*
- Unsupervised Learning, Recommenders, Reinforcement Learning — *Coursera / Stanford & DeepLearning.AI (2023)*
- Django Web Development – *Mindrisers Institute of Technology (2023)*
- MERN Stack Development – *The British College (2023)*